

Machine Learning Final Project – Part I

Predicting Chronic Kidney Disease

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Problem Statement

Chronic Kidney Disease (CKD) is a condition that can become very serious if it is not diagnosed and treated early. Doctors use different types of patient information, such as laboratory results, medical history, and other clinical measurements, to determine whether a patient has CKD. Reviewing all of this information manually can take a significant amount of time, especially when there are many patient records.

This project addresses a binary classification problem. The goal is to develop a machine learning model that predicts whether a patient has Chronic Kidney Disease based on the available clinical and laboratory data. A predictive model could help healthcare providers identify patients who may be at risk and support earlier diagnosis and treatment decisions.

Project Objective

The objective of this project is to prepare and analyze the dataset for the development of a machine learning model that predicts whether a patient has Chronic Kidney Disease.

This first part of the project focuses on exploring the dataset, reviewing its quality, and preparing the data before moving on to model development.

Success Measures

The project will be considered successful if the machine learning model can accurately classify patients as having or not having Chronic Kidney Disease. Although the model is not developed during this phase, future performance will be evaluated using classification metrics such as accuracy, precision, recall, and F1-score. Success for this phase also includes identifying the target variable, evaluating data quality, documenting the exploration process, and preparing the dataset for future model development.

Assumptions

Assumption	How it will be verified
The dataset represents patients with and without CKD.	Review the dataset for missing values, consistency, and completeness during data exploration.
CKD_class correctly represents the diagnosis.	Compare CKD_class with the classification column to confirm they represent the same outcome.
Missing values can be handled appropriately.	Evaluate the amount of missing data and apply appropriate preprocessing techniques before model training.
The selected features provide useful information.	Review feature relationships during data exploration and later evaluate their contribution during model development.

Target Variable

The dataset contains two columns related to the kidney disease diagnosis: classification and CKD_class. Since both columns represent the same diagnosis, CKD_class was selected as the target variable because it is already coded as 0 and 1, making it more appropriate for machine learning. The classification column was removed from the input features to avoid using duplicate information during model development.

Data Quality Assessment

The dataset contains 400 patient records and 27 variables. During the exploration phase, several columns were identified with missing values, while the target variable (CKD_class) contains complete information for all records. Summary statistics, missing value analysis, histograms, and a correlation matrix were used to better understand the dataset. A working copy of the data was created, unnecessary columns were removed, the target variable was identified, and the dataset was divided into training and testing datasets.

Conclusion

This first phase of the project established a foundation for the remaining work by framing the problem, exploring the dataset, and completing the initial data preparation steps. The dataset is suitable for continued preprocessing before developing and evaluating the machine learning model.